

Groundwater Flow in a Cofferd Dam

In this tutorial, finite element groundwater seepage analysis is used to determine the quantity of seepage entering a cofferdam. The problem is solved in three stages as follows:

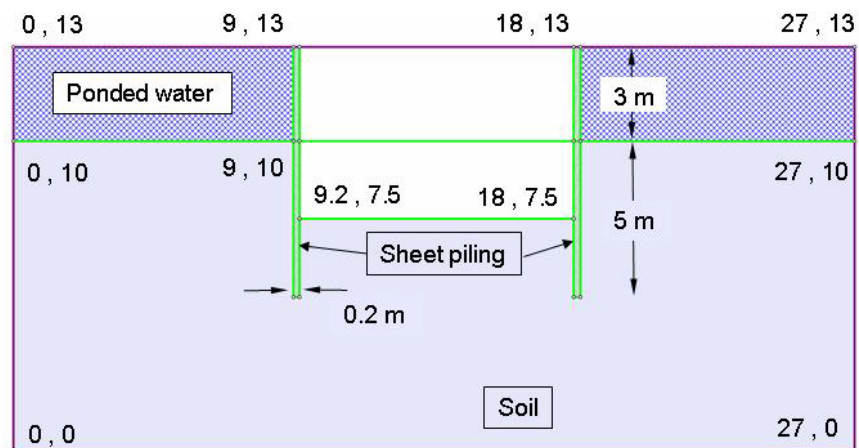
1. Installation of sheet pilings to hold back water
2. Excavation of soil inside the dam
3. Pumping of water inside the dam

The example is based on problem 2.4 from Craig (1997). The problem is constructed and solved entirely with *Phase2*.

Topics Covered

- Seepage analysis
- Renaming stages
- Multiple materials
- Relative coordinates
- Boundary copying
- Discharge sections
- Groundwater only calculation
- Flownets

Geometry

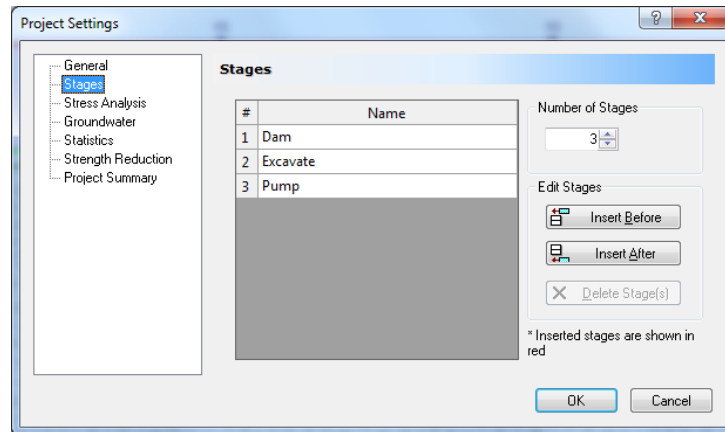


Model

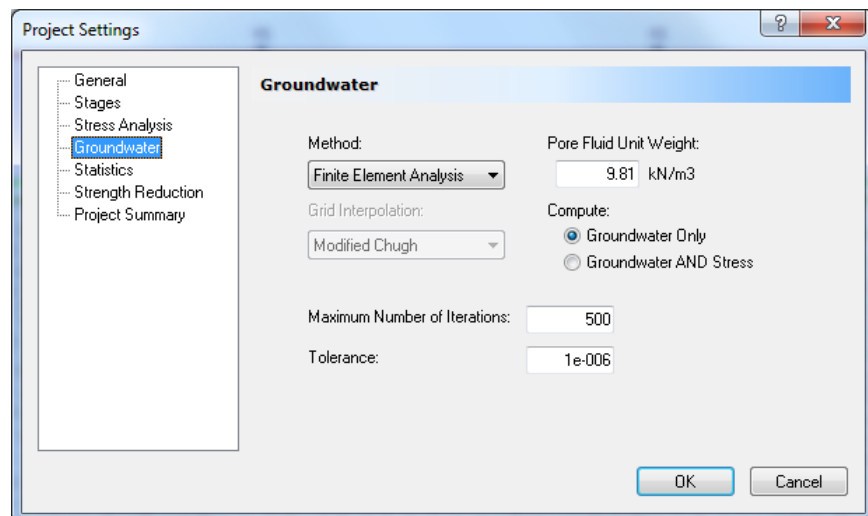
Start the *Phase2* Model program.

Project Settings

Open the Project Settings dialog from the Analysis menu and make sure the General tab is selected. Define the units as being “Metric, stress as kPa”. Change the number of stages to 3. Name the first stage “Dam”, the second stage “Excavate” and the third stage “Pump” as shown.



Click OK to close the Rename Stages dialog. Back in the Project Settings dialog, select the Groundwater tab. Under Method choose Finite Element Analysis. This enables steady state finite element analysis of groundwater flow. Under Compute choose Groundwater Only since we are not interested in solid deformations and stresses in this tutorial. Close the Project Settings dialog by pressing the OK button.



Boundaries

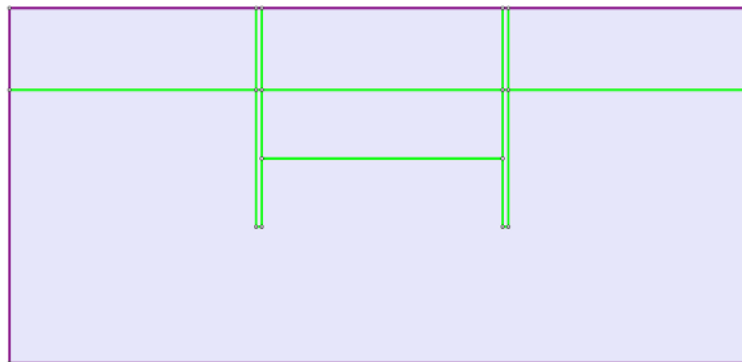
First add an external boundary. Select the **Add External** option in the Boundaries menu and enter the four corner coordinates shown in the figure at the beginning of this tutorial.

Now we need to add boundaries to separate the ponded water from the soil and to delineate the impermeable sheet piling. First draw a horizontal material boundary across the model by choosing Add Material from the Boundaries menu. Enter the coordinates (0,10) and (27,10). Hit Enter to finish.

The sheet piling is assumed to be very thin. To create the boundary for the sheet piling therefore, it is easier to enter relative distances instead of absolute coordinates. Go to the Boundaries menu and select Add Material. Enter the top left coordinate for the right sheet piling (18,13) and hit enter. Now you can enter a relative distance from this point for the next point. Enter @ 0.2 , 0 in the prompt line and hit enter. This will put the next point 0.2 m away in the x direction and 0 m in the y direction. Now enter @ 0 , -8 to plot the next point 8 m below. Now enter @ -0.2 , 0 and finish with @ 0 , 8 to close the boundary.

To create the left sheet pile, simply right click on the right sheet pile, select Copy Boundary, and enter the relative coordinates @ -9 , 0. Hit Enter.

Finally, enter another horizontal material boundary to delineate the water level inside the dam for stage 2 (excavation). Select Add Material from the Boundaries menu and enter the coordinates 9.2 , 7.5 and 18,7.5. Hit Enter to finish. Your model should now look like this:



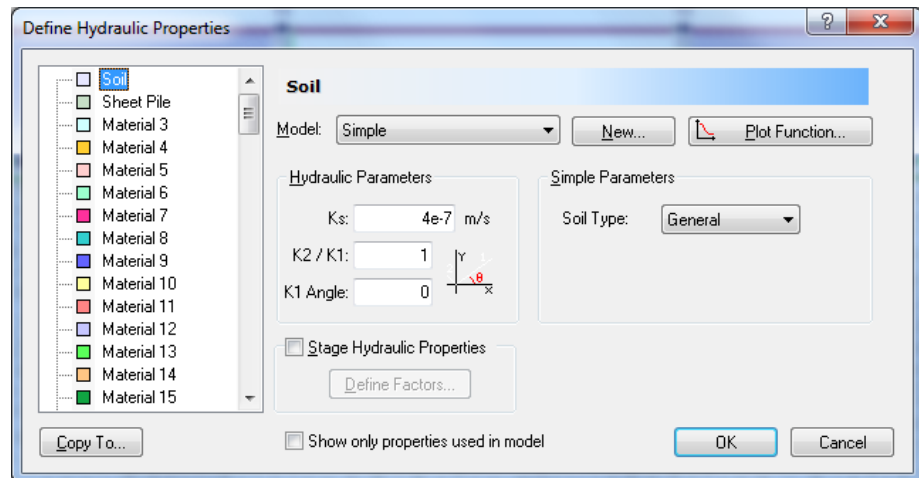
TIP: When you need to enter a lot of complicated geometry to your model, it can be useful to see the distance between points and the angles between lines. You can do this by going to File and choosing Preferences and turning on the option *Show segment length, radius, angles when adding geometry*.

Material Properties

Select Define Materials from the Properties menu. You will now see all of the default material properties for Material 1. In this tutorial we don't care about the strength and stiffness of the solid material therefore leave all the default values. Change the name of Material 1 to Soil. Now click on the Material 2 tab. Change the name of Material 2 to Sheet Pile. Click OK to close the dialog.

We now need to define the fluid flow properties of the soil. To do this, go to the Properties menu and choose Define Hydraulic. Click on the Soil tab at the top of the dialog. Enter $4\text{e-}7$ for Ks. Ks is the saturated permeability in m/s (also called hydraulic conductivity). You may specify anisotropic permeability by specifying $K2/K1 \neq 1$ and an angle to indicate the directionality. However we will assume isotropic permeability so do not change the default values.

The **Model** option at the top of the dialog refers to the function used to calculate the permeability in the unsaturated zone as a function of matric suction. Different models may be chosen, including a user-defined model. However we will use the default Simple option. See the *Phase2* Help for more information on permeability models. Your dialog should now look like this.

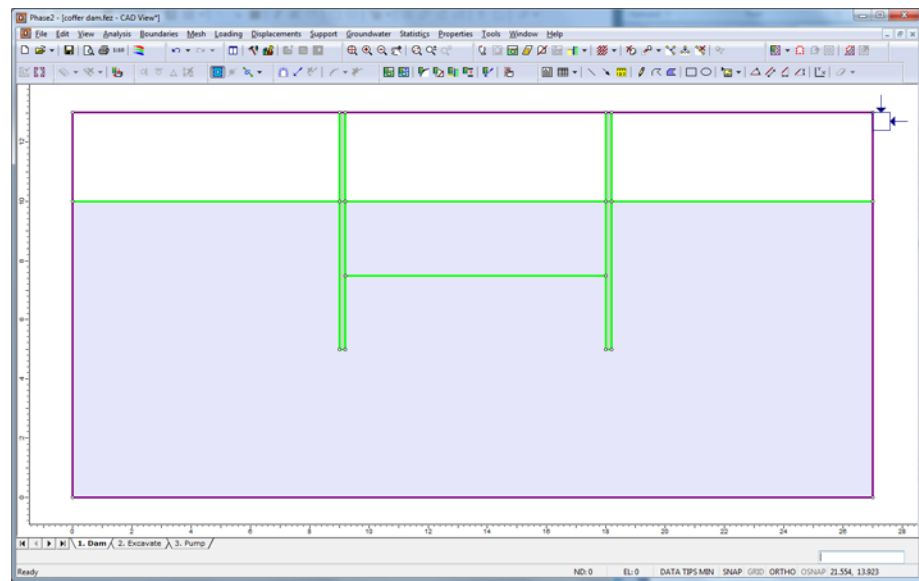


Now select the Sheet Pile tab. The sheet piling is assumed to be essentially impermeable. We wish to set the permeability to a very low value, however we cannot choose 0 since this will lead to numerical instability. Therefore set the permeability, Ks, to $1\text{e-}20$. Click OK to close the window.

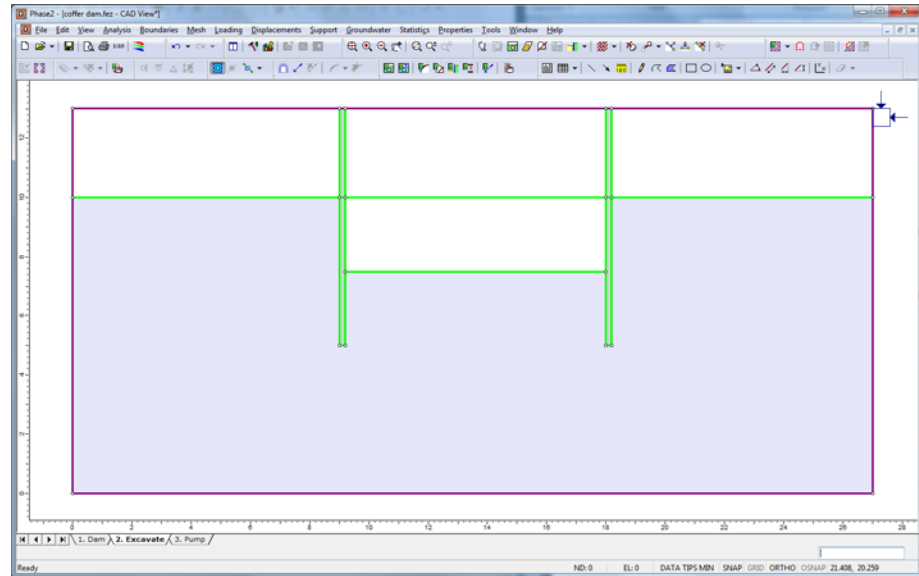
Assign Material Properties

By default, the entire model is assigned the properties of Soil (material 1). In our model, the top left and top right sections will represent ponded water and the top middle section will be empty space. We therefore need to remove the soil material from these areas. Ensure that you are looking at Stage 1 (Dam). Select Assign Properties from the Properties menu. Click on the Excavate button and then select the top left, top middle and top right sections of the model. Do not close the dialog.

We now wish to assign the Sheet Pile properties to the sheet pilings. Select Sheet Pile from the Assign dialog and click inside the 4 sections representing the 2 sheet pilings. Do not close the dialog. Your model for Stage 1 should now look like this.

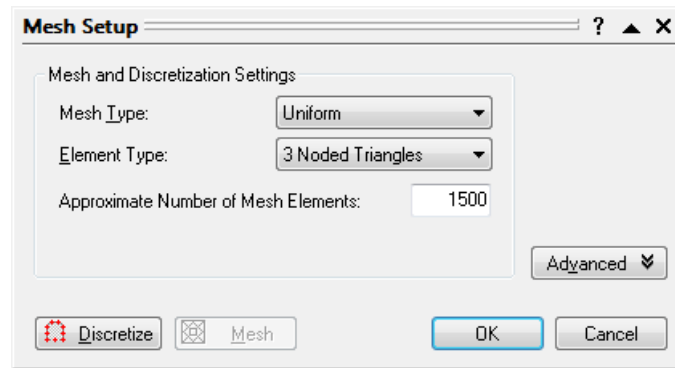


Now click on the Stage 2 (Excavate) tab at the bottom of the *Phase2* window. Click the excavate button in the Assign dialog and then click in the middle section of the model to excavate the area between the sheet pilings. Close the assign dialog. Your model for Stage 2 (Excavate) should appear as below.

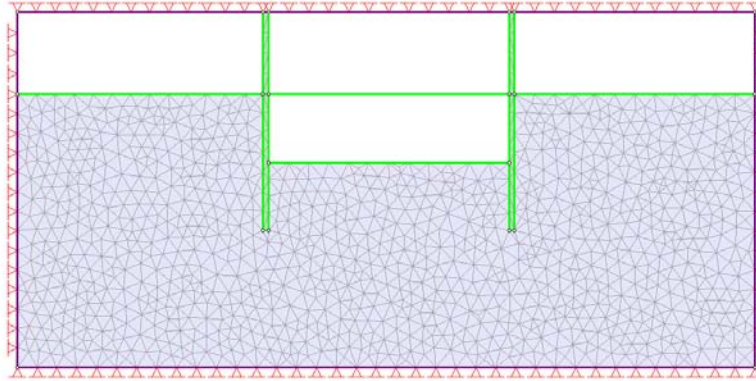


Mesh

Now generate the finite element mesh. Select the **Mesh Setup** option in the **Mesh** menu. Change the Mesh Type to Uniform. Leave the default element type (3 Noded Triangles) and the number of elements (1500). Click the Discretize button followed by the Mesh button.



Close the Mesh Setup dialog by selecting the OK button. Your model should now appear as shown.

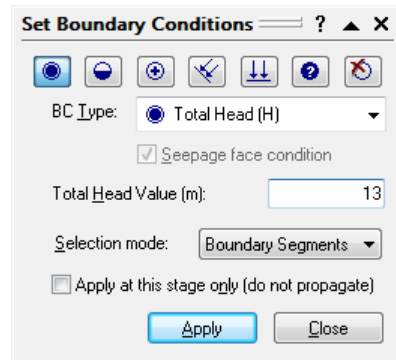


Boundary Conditions



The model shows the default boundary conditions for the solid soil (all external boundaries fixed in the x and y directions). Since we are only performing a groundwater analysis, we don't care about the stress and displacement boundary conditions. To set the boundary conditions for the groundwater analysis, select Set Boundary Conditions from the toolbar or the Groundwater menu.

Make sure you are looking at Stage 1 (Dam). For all stages we wish to simulate ponded water to the left and right of the sheet piling. The elevation of the top of the sheet piling is 13 m. Therefore we will set the total head for these boundaries to 13 m. To do this, choose Set Boundary Conditions from the Groundwater menu. For BC Type choose Total Head. Enter a Total Head Value of 13.



Now select the four boundary segments that enclose the ponded water:

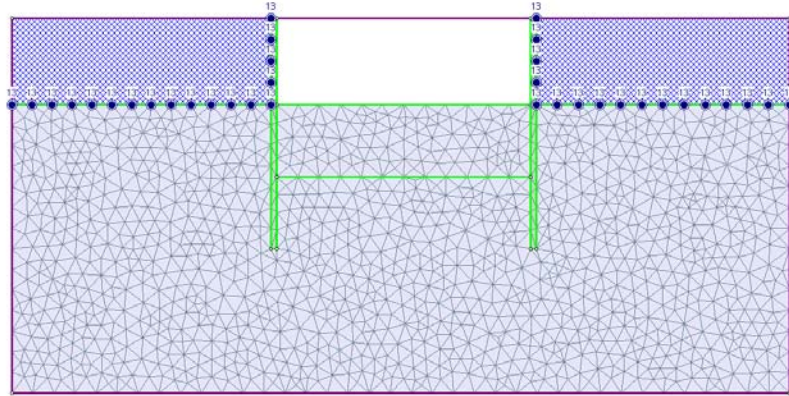
Line 1: from (0,10) to (9,10)

Line 2: from (9,10) to (9,13)

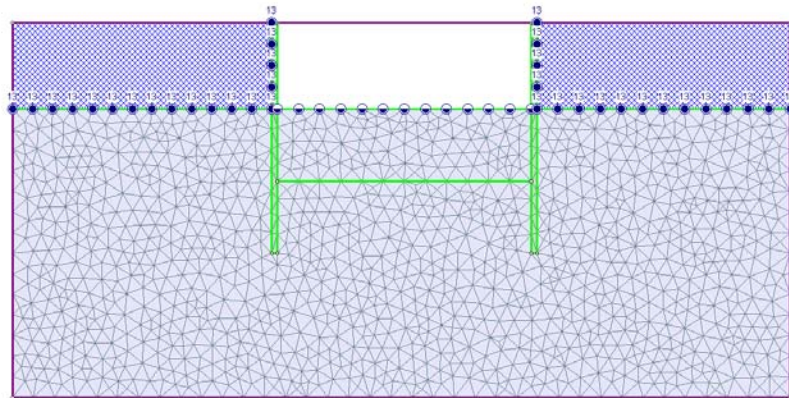
Line 3: from (18.2,10) to (18.2,13)

Line 4: from (18.2,10) to 27,10)

Click Apply and your model should look like this.



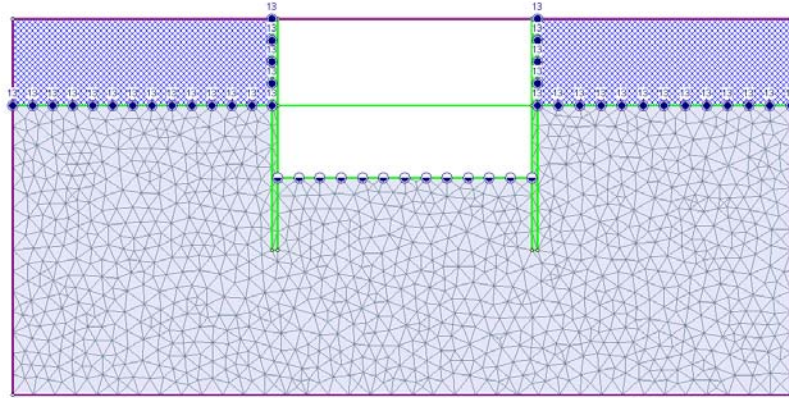
The soil surface inside the coffer dam has zero pore pressure (it is at atmospheric pressure). Therefore we need to set the pressure on this surface to zero. In the Set Boundary Condition Dialog, choose Zero Pressure for the BC Type. Click on the ground surface between the pilings and hit Enter (or right click and choose Done Selection). Now close the dialog box. Your model will appear as shown.



Boundary conditions applied at Stage 1.

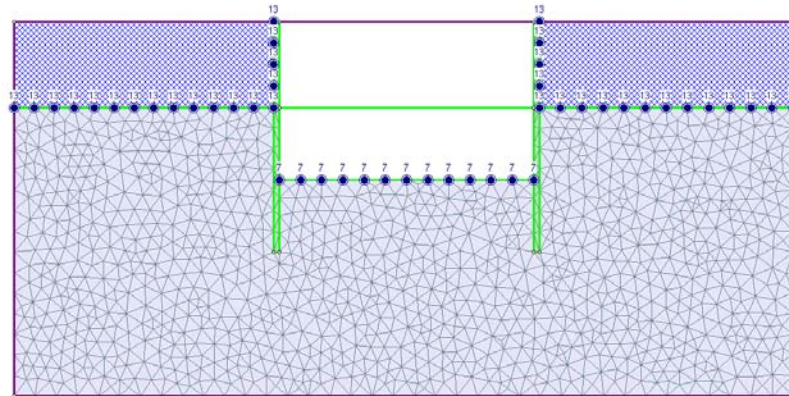
TIP: you can also right-click on a boundary to define its boundary conditions.

Now select the Stage2 (Excavate) tab at the bottom of the window. Set the pore pressure for the top of the soil layer to zero as you did for Stage 1. You may also remove the zero pressure BC for the boundary where the soil used to be in Stage 1 (although this is not necessary).



Boundary conditions applied at Stage 2.

Finally, select Stage 3 (Pump). In this stage we wish to simulate pumping such that the water table will be below the surface. To do this set the Total Head for the soil surface between the pilings to 7 m. This is slightly below the actual elevation of 7.5 m. The model should now appear as shown.



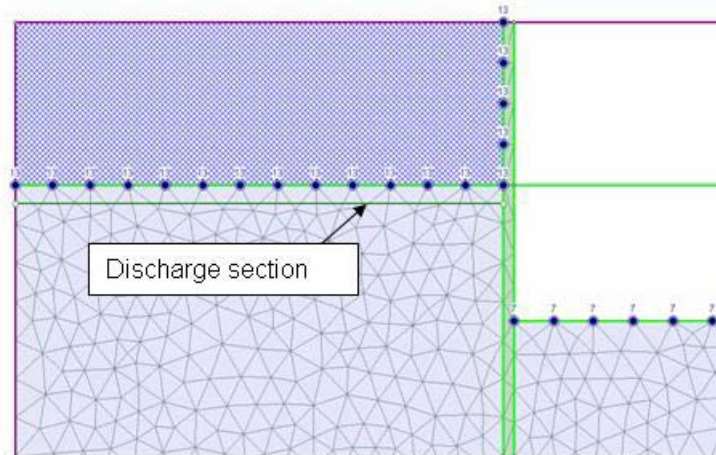
Boundary conditions applied at Stage 3.

We are now done applying the boundary conditions so close the Set Boundary Conditions Dialog.

Discharge Sections

If we wish to calculate flow quantities, this is done by defining a Discharge Section. A Discharge Section in *Phase2* is a user-defined line segment, through which the steady state, volumetric flow rate, normal to the discharge section, will be calculated during a groundwater seepage analysis.

We wish to add discharge sections at the soil surface. To do this, choose **Add Discharge Section** from the Groundwater menu. Enter a start point on the left boundary just below the ponded water. Add a finish point on the left edge of the left sheet piling just below the ponded water. After the second point is entered, the discharge section will be added to the model, and you will automatically exit the Add Discharge Section option. You can enter the coordinates using the keyboard but it is easier to just click on the model since the cursor will snap to the boundaries (if the cursor does not snap to the boundaries go to the View menu, select Snap and ensure all of the options are selected). The discharge section is displayed as a green line, with small circles marking the endpoints as shown. The value of the flow rate across this line will be displayed in the *Phase2* Interpret program, when you view the analysis results.

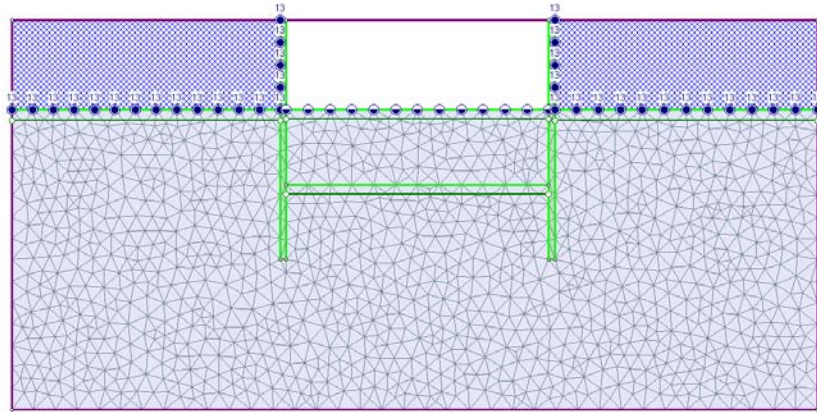


TIP: you can delete a discharge section by right-clicking on it and choosing Delete Discharge Section.

Now perform the same steps to set three more discharge sections: one below the soil surface on the other side under the ponded water, one below the soil surface between the sheet pilings in Stage 1 and one below the soil surface between the sheet pilings in Stage 2.

NOTE: It doesn't matter at what stage you add the discharge sections. They will persist throughout the whole model.

Your final model should now appear as shown.



Cofferdam model – Stage 1 (Dam)

You have completed the definition of the model. Save the model using the Save option in the File menu.

Compute

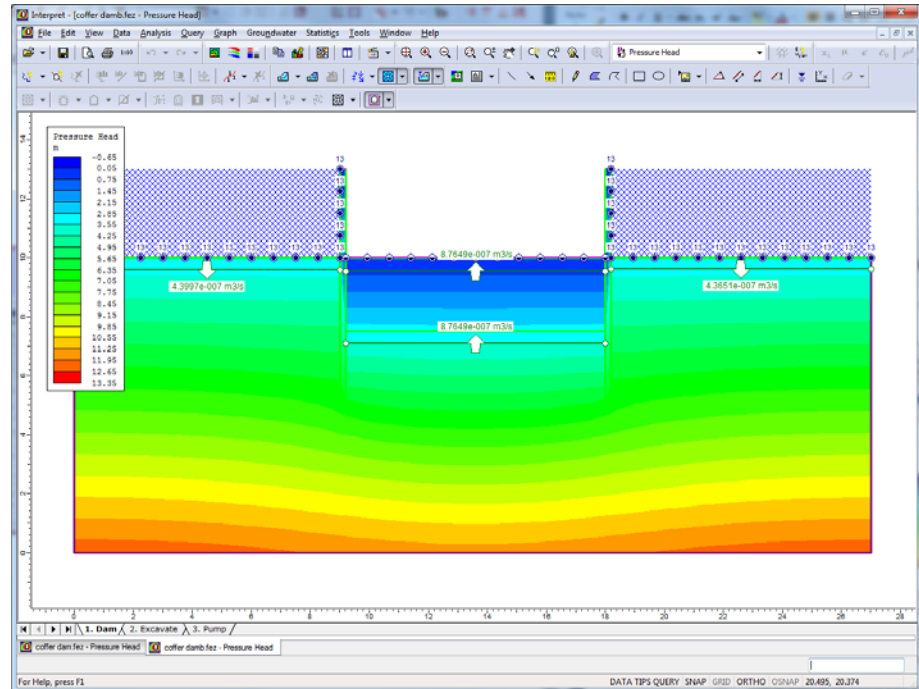


Recall that the model is set up for groundwater computation only. Therefore you must run the model using the **Compute (groundwater only)** option in the **Groundwater** menu (or click the Compute groundwater only button in the toolbar). The analysis should take a few seconds to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

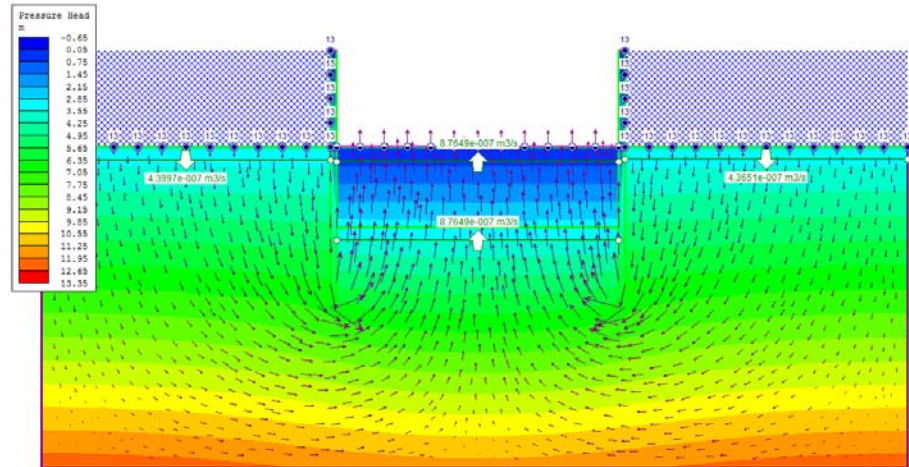
After you select the Interpret option, the Interpret program starts and reads the results of the analysis. The following screen is displayed showing the pressure head results for Stage 1 (Dam).



You can also see the volumetric flow rate and direction through each of the discharge sections. As you would expect, the water is flowing down from the ponded water and up into the dam. The sum of the volumetric downwards flow is equal to the volumetric upwards flow between the sheet pilings.

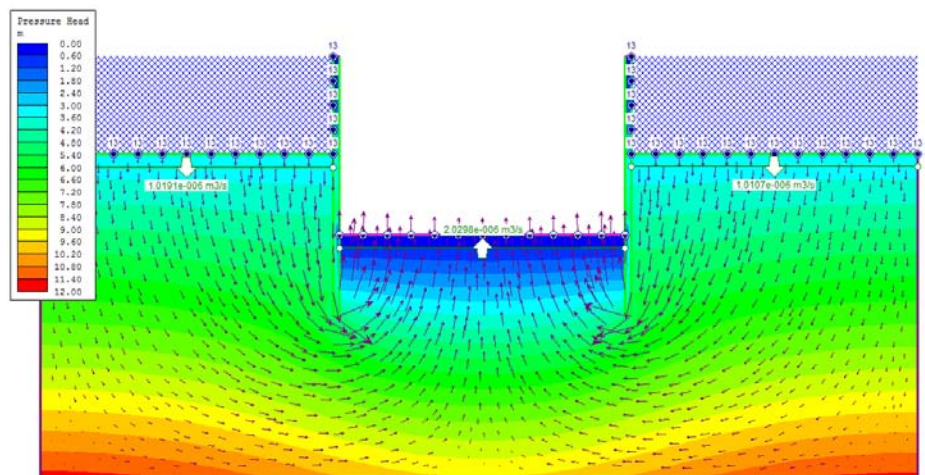


To see the magnitude and direction of flow throughout the model, plot the Flow Vectors by clicking the Flow Vectors button. It is clear that the groundwater is flowing around the impermeable sheet pilings with high flow rates directly below the pilings.



Pressure head and flow vectors for stage 1

Now click on the Stage 2 tab (Excavate). You can now see the flow conditions after some soil has been excavated. This geometry corresponds to Problem 2.4 in Craig (1997). This problem asks for the quantity of seepage entering the cofferdam. From the figure below, the volumetric flow into the dam is $2.0234\text{e-}6 \text{ m}^3/\text{s}$. The value given in Craig (1997) is $2.0\text{e-}6 \text{ m}^3/\text{s}$.

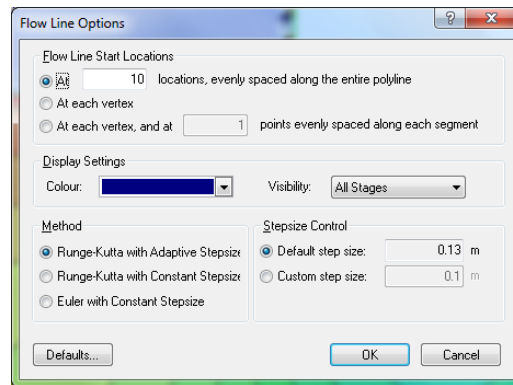


Pressure head and flow vectors for stage 2

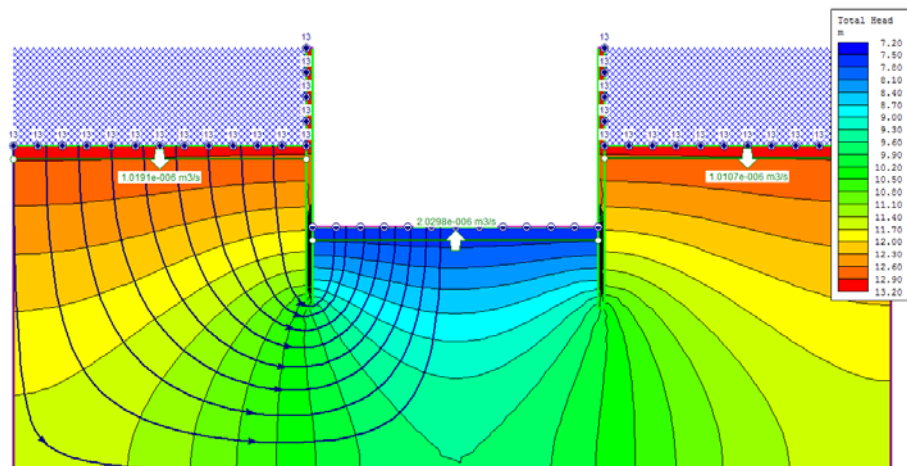
TIP: you can hide the display of the discharge section that is no longer in the soil by right clicking on it and selecting Hide This Discharge Section.

The problem also asks for a flow net to be constructed. This can also be done with *Phase2*. First turn off the flow vectors by pressing the flow vector button again. Now change the quantity being plotted from Pressure Head to Total Head using the drop down menu on the tool bar. Now right-click on the model and select Contour Options. Under Mode select Filled (with lines) and then select Done. You will now see the equipotential lines of the flownet.

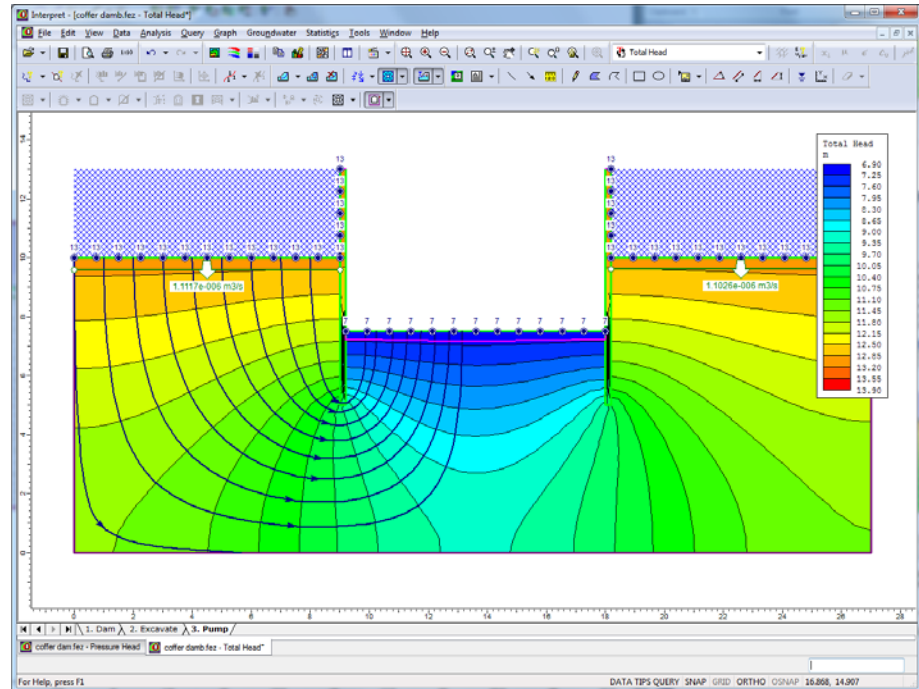
To plot the flow lines, go the Groundwater menu and select Add Multiple Flow Lines. Select the top left corner of the soil as the first point (you may need to move the legend out of the way prior to this). If the cursor does not snap to the node point go to the View menu, select Snap and ensure that all snap options are turned on. Now move horizontally until you intersect the sheet piling and click to establish the second point. Hit enter to finish. You will now see the Flow Lines Options dialog. Here you can choose how many flow lines you wish to plot. Under Flow Line Start Locations select the first option and leave the default value (10 locations, evenly spaced along the polyline). Under Flow Line Method select the first option and leave the default value (10 locations, evenly spaced along the polyline).



Click OK to close the dialog. You will now see 10 flow lines plotted as shown. To complete the flownet you could repeat these steps for the right side of the model.



Now select the Stage 3 tab (Pump). For this stage, pumping was simulated by applying a total head lower than the elevation. You can see that the volumetric discharge at the bottom of the dam is higher than in Stage 2. You can also see that the water table has been lowered. The water table is shown as a pink line (your water table line may be obscured by the green discharge line. To hide the discharge line, right click on it and choose Hide This Discharge Section).



This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.

References

Craig, R.F., 1997. *Soil Mechanics*, Spon Press, London and New York, 485 pp.